

SYED MOHAMMED MIFTAHUDDIN MARIF

SECURITY ANALYST

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SUMMARY

Security-focused Software Engineer pursuing an M.E. in Cyber Security at Osmania University, with hands-on experience in Attack Surface Management (ASM), vulnerability assessment, network analysis, and secure software development. Experienced in identifying exposed assets, validating security weaknesses, analyzing network traffic, and documenting findings through professional security assessment reports. Strong foundation in application, network, cloud, and endpoint security, with practical experience in AWS VPC, IAM, security controls, and infrastructure configuration. Skilled in Wireshark, Burp Suite, Nmap, Maltego, and Python, with additional experience in Java, Spring Boot, REST APIs, and microservices. Interested in Security Analyst, SOC, Cloud Security, and Application Security roles.

TECHNICAL SKILLS

Security Domains: Attack Surface Management (ASM), Vulnerability Assessment, Penetration Testing (PTVA), Web Application Security, Network Security, Malware Analysis, Digital Forensics.

Security Tools: Wireshark, Burp Suite, Nmap, Maltego, Metasploit, MXToolbox, dig, Postman, Process Hacker, Hydra, FTKimager, Volatility3, Autopsy.

Security Frameworks & Concepts: MITRE ATT&CK Framework, Cyber Kill Chain, OWASP Top 10, OWASP WSTG, CVSS, Threat Modeling, Secure Coding Practices, Authentication & Authorization, Cryptography Fundamentals, Risk Assessment

Networking: TCP/IP, DNS, HTTP/HTTPS, Routing & Switching, Packet Analysis

Cloud & Systems: AWS (EC2, VPC, Security Groups, Application Load Balancer, RDS, IAM, S3), Azure, Linux, Windows, Docker, Git, GitHub

Programming & Scripting: Python, Java, JavaScript (ES6+), C/C++

Web & Application Development: Spring Boot, Spring Framework, Node.js, Express.js, RESTful APIs, Microservices, OAuth 2.0, JWT Authentication

Databases: PostgreSQL, MySQL, MongoDB

INTERNSHIPS EXPERIENCE

HulkHire Tech, Hyderabad | Java Developer Intern | Apr 2025 – Aug 2025

- Developed and integrated PayPal Checkout using Java Spring Boot in a microservices architecture on AWS
- Implemented end-to-end payment lifecycle & reconciliation, ensuring zero unsettled transactions
- Secured APIs using OAuth 2.0 and improved performance with Redis caching
- Designed backend using Factory & Builder patterns with centralized exception handling
- Configured AWS VPC, subnets, and security groups; deployed services on EC2 with Application Load Balancer
- Managed RDS databases and AWS Secrets Manager for secure credential handling
- Improved fault tolerance using circuit breaker pattern and monitoring via Micrometer
- Achieved high test coverage using JUnit & Mockito

Anudip Foundation | Computer Hardware Engineer Trainee | OCT 2021- Dec 2021

- Performed system assembly, hardware diagnostics, maintenance, and troubleshooting across desktops, workstations, and peripheral devices.
- Configured and maintained operating systems, device drivers, BIOS/UEFI settings, storage, memory, and system-level components.
- Configured basic LAN connectivity, IP addressing, network adapters, routers/switches, and peripheral network devices.
- Troubleshoot TCP/IP connectivity, DNS, DHCP, network configuration, and common endpoint communication issues.
- Applied basic endpoint security practices, including system hardening, access control, software configuration, and security updates.
- Diagnosed hardware, operating system, and network issues using systematic troubleshooting and root-cause analysis.

EDUCATION

Masters of Engineering in Cyber Security | Nov 2025 – Sep 2027

University College of Engineering - Osmania University

Bachelor of Engineering in Computer Science and Engineering | Nov 2021 – Dec 2024

Muffakham Jah College of Engineering and Technology – Affiliated by Osmania University

Diploma in Computer Engineering | Jun 2018 – Sep 2021

Quli Qutub Shah Polytechnic College – State Board of Technical Education.

ACHIEVEMENTS

- **Qualified GATE 2026 (CSE) – Score: 550 | Rank: 8810**
- **Secured 55 Rank in TS PGECET**

PROJECTS

AI-Based SSH Intrusion Detection System | Python, scikit-learn, pandas, Flask

- Designed a hybrid intrusion detection system combining unsupervised Isolation Forest anomaly detection with a rule-based engine to flag SSH brute-force, credential-stuffing, and spoofing attacks with zero labeled training data.
- Engineered a behavioral feature pipeline extracting 11 metrics (failed-login counts, username diversity, auth-failure rates) per source IP over sliding 5-minute windows from 2,000+ raw server log entries.
- Built a real-time monitoring service in Flask with a background file-watcher thread, REST API, and live-polling dashboard that surfaces new alerts within 4 seconds of detection.
- Implemented severity-ranked, evidence-backed alerting that fuses ML anomaly scores with interpretable rule triggers, surfacing 21 ranked alerts across 13 of 30 monitored IPs on the OpenSSH LogHub-2k benchmark.
- Authored a full technical research paper and presentation deck covering system architecture, related work, and experimental evaluation.

AWS VPC Network & Cloud Security Lab | AWS, VPC, IAM, EC2, Security Groups, NACL, Bastion Host

- Designed and deployed a custom AWS VPC with public and private subnets, route tables, Internet Gateway, and controlled network traffic flows.
- Configured Security Groups and Network ACLs to implement inbound/outbound traffic controls and apply defense-in-depth network security.
- Deployed an EC2 Bastion Host as a controlled administrative entry point for accessing resources in private subnets.
- Implemented IAM users, groups, roles, and policies following least-privilege principles for controlled access to AWS resources.
- Analyzed AWS networking concepts including CIDR, IP addressing, routing, subnetting, NAT, DNS, and inter-subnet communication through hands-on configuration.
- Practiced cloud security fundamentals including network segmentation, access control, secure administrative access, and restriction of unnecessary network exposure.

College Website ASM and VAPT | Pragya Cyber Project.

- Conduct Asset Surface Management (ASM) activities to identify internet-facing assets, exposed services, and potential attack vectors.
- Perform vulnerability assessments and limited penetration testing (VAPT/PTVA) on web applications and external infrastructure.
- Analyze findings, validate vulnerabilities, and assess security risks based on impact and exploitability.
- Prepare detailed technical reports documenting vulnerabilities, evidence, risk ratings, and remediation recommendations.
- Utilize security tools such as Nmap, Burp Suite, Wireshark, and OSINT techniques during assessments.
- Collaborate with senior security professionals to improve assessment methodologies and reporting quality

Smart Irrigation & Remote Surveillance System | ESP8266, ESP32-CAM, IoT, Networking

- Developed an IoT-based smart irrigation system using ESP8266, DHT11, soil moisture, and rain sensors for automated environmental monitoring and watering control.
- Built a web-based dashboard enabling real-time sensor monitoring, manual control, and scheduled irrigation, with configuration persistence using EEPROM.
- Enabled remote internet access using No-IP dynamic DNS and router port forwarding, connecting the local IoT system to an externally accessible endpoint.
- Integrated ESP32-CAM for live video surveillance and visual monitoring alongside environmental sensor data.
- Configured and troubleshoot LAN connectivity, IP addressing, port forwarding, and device-to-network communication for reliable remote access.
- Designed the system with modular architecture for future cloud integration, centralized monitoring, and AI-based irrigation optimization.

CERTIFICATES

Honeywell Certified Cybersecurity Analyst 